CACTUS FALL 2026 · WEEK 7 · SESSION 1 · LEARNING SESSION

Applying statistics to social media data

Plant an answer, run the method, check before you trust it.

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THE IDEA

Plant an answer, then check

Week 3

We planted an effect in simulated data and found it. We knew the truth, so we could judge the method.

Week 6

We measured an AI against people. We needed a reference to know whether to trust it.

Today

Do the same for any method: plant the answer, run the method, compare. Then see what messy data does to it.

Social media data adds one more problem: the rows are not independent. That is most of today.

KNOWN-ANSWER TESTS

Simulate, run, compare

```
effect = 0.5 # you choose the truth
d = simulate(effect) # make data with that truth planted
est = fit(d) # run the method you plan to use
print(est, "vs", effect) # compare: how close, and how often?
```

Run it once and you see one result. Run it **100 times** and you see how often the method gets it right, and how far off it can be.

THE LIMITS

What it cannot tell you

What it shows

The method works when the data behave the way you simulated.

What it hides

Real data has bots, repeats, groups, and small samples that you did not plant.

What to do

Plant the mess on purpose, one problem at a time, and watch the answer move.

A method that passes a clean test and fails a messy one is telling you where your real data will hurt.

ROWS THAT COME IN GROUPS

Twelve comments, three posts

Post	Warmth of four comments (invented)
Post A	8.1 · 7.9 · 8.4 · 8.0
Post B	4.2 · 4.0 · 4.5 · 3.9
Post C	6.1 · 6.3 · 5.8 · 6.0

Comments under the same post are alike, because they share the post. **How many independent chances** do you have to learn what posts do: 12, or 3? Now imagine 3,000 comments under 80 posts.

THE FIX

Standard errors that notice groups

Naive

Every comment is one independent row. Looks very certain, and is too certain for anything the post decides.

Clustered

Comments under a post count as one cluster. The estimate does not change. The uncertainty gets honest.

Mixed model

A model with a bump for each post. One line of code, the same idea.

Which variables are hurt most? The ones that belong to the **post**. A variable that changes from comment to comment is fine.

COUNTS AND EXPOSURE

Comments per follower, not comments

Post (invented)	Followers	Comments	Per 1,000 followers
X	1,000	10	10
Y	10,000	30	3

Y has three times the comments and less than a third of the rate. A count needs a **denominator**. A model tells you the denominator with an **offset**.

A CHECK THAT ALWAYS WORKS

Permutation: what chance looks like

1. Shuffle

Mix up the labels so any pattern is down to chance.

2. Recompute

Calculate your statistic on the shuffled data. Do it 1,000 times.

3. Compare

Where does your real number fall among the chance numbers?

Shuffle at the level the label **belongs to**. A label that belongs to a post is shuffled across posts, not across comments. It works when there is no formula, such as a network statistic.

SESSION 2

Three tracks

A. Networks

Who talks to whom. Centrality, communities, bots. Check against shuffled networks.

B. Images and memes

Fingerprints, near-duplicates, templates. The same image counted many times.

C. Mock feed or experiment

Conditions, a manipulation check, an interaction. Who you drop matters.

In each, the data are invented and the answer is planted. Choose at the end of the coding session today.

WHAT TO DO NEXT

Before the coding session

Practice page

Plant It, Find It, about 20 minutes. It ends with a planted answer in Colab.

Required reading

Textbook Chapter 8, sections 8.1, 8.2 and 8.5. Read them with this question: what if my rows come in groups?

Data

All the exercise data is invented and public. No class drive this week.

In the coding session: plant an answer, see naive versus clustered, model counts with an exposure, find three mistakes in AI-written code, and try a permutation test. Choose your track at the end.

QUESTIONS

Where do your rows come in groups?

Think of your own project. Comments in posts, posts by one account, people in one group. Write one sentence in your project card.